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WorkPulse

Attendance & Leave Management System

System Documentation

Version 1.0 · April 2026

1. Project Overview

WorkPulse is a full-stack web-based Attendance and Leave Management System designed for companies with multiple branches. It provides real-time attendance tracking with GPS geofencing, leave application and approval workflows, automated email notifications, overtime management, and comprehensive reporting — all accessible through a single-page web application.

■ Designed for Indonesian companies: includes WIB/WITA/WIT timezone support, Indonesia national public holidays for 2025–2026, and multi-branch geofencing.

2. Technology Stack

Layer	Technology	Version	Purpose
Backend	Python / Flask	3.x / 3.0.3	REST API server
Database	PostgreSQL	15+	Persistent data storage
DB Driver	psycopg2-binary	2.9.9	Python → PostgreSQL adapter
Web Server	Gunicorn	22.0.0	Production WSGI server
Frontend	HTML5 / CSS3 / Vanilla JS	ES2020+	Single-page application
PDF Export	Browser Print API	Native	Client-side PDF generation
Excel Export	SheetJS (xlsx)	0.18.5	Client-side .xlsx generation
Email	Python smtplib	stdlib	SMTP email via Gmail
Geofencing	Haversine formula	—	GPS distance calculation
Hosting	Railway.app	Cloud	Web service + PostgreSQL DB
Auth	Flask Sessions + SHA-256	—	Server-side session auth

3. System Architecture

WorkPulse follows a classic three-tier architecture with a single-page frontend, a RESTful Flask API backend, and a PostgreSQL database. All PDF and Excel generation happens client-side in the browser to minimise server load.

Layer	Component	Responsibility
Presentation	index.html + app.js + style.css	SPA rendering, user interaction, client-side PDF/Excel
Application	app.py (Flask)	API routing, business logic, geofence checks, email
Data	PostgreSQL on Railway	Persistent storage for all users, attendance, leave, setting
External	Gmail SMTP	Leave approval & missing clock-out email notifications
External	Browser Geolocation API	GPS coordinates for punch-in/out geofencing

Key Design Decisions

- **Stateless REST API:** All state stored in PostgreSQL; Flask session only holds user ID and role.
- **Client-side exports:** PDF uses `window.print()`, Excel uses SheetJS — no server dependencies needed.
- **Auto DB migration:** `init_db()` runs on every startup and applies schema changes safely.
- **Timezone-aware:** All server timestamps use zoneinfo with TZ environment variable (default: Asia/Jakarta).
- **Geofence flexibility:** Each branch has its own coordinates and radius; employees assigned per branch.

4. Database Schema

users

Stores all employees, managers, and HR admins.

Column	Type	Description
id	SERIAL PK	Auto-increment primary key
employee_id	TEXT UNIQUE	Company employee ID code
first_name / last_name / name	TEXT	Split and combined name fields
email	TEXT UNIQUE	Login email address
password	TEXT	SHA-256 hashed password
role	TEXT	employee manager hr_admin
department	TEXT	Department name
branch_id	INTEGER FK	Assigned office branch for geofencing
manager_id	INTEGER FK	Self-referencing supervisor FK
shift_start / shift_end	TEXT	Shift hours in HH:MM format
reset_token / reset_expires	TEXT	Password reset token and expiry

branches

Office locations used for geofence validation.

Column	Type	Description
id	SERIAL PK	
name	TEXT	Branch display name
address	TEXT	Physical address
latitude / longitude	DOUBLE PRECISION	GPS coordinates
radius_m	INTEGER	Allowed radius in metres (default 200)

attendance

Daily clock-in and clock-out records.

Column	Type	Description
id	SERIAL PK	
user_id	INTEGER FK	References users.id
date	TEXT	YYYY-MM-DD
punch_in / punch_out	TEXT	HH:MM:SS clock times
status	TEXT	ontime late absent leave
lat_in/lon_in/lat_out/lon_out	DOUBLE PRECISION	GPS coordinates at punch
geo_in / geo_out	TEXT	Human-readable distance label
notes	TEXT	JSON blob for overtime metadata

leave_types

Configurable leave categories.

Column	Type	Description
id	SERIAL PK	
name	TEXT	E.g. Annual Leave, Sick Leave
max_days	INTEGER	Maximum days allowed per year

leave_balances

Per-employee per-year leave allowance tracking.

Column	Type	Description
user_id + leave_type_id + year	UNIQUE	Composite unique constraint
total_days	INTEGER	Allocated days for the year
used_days	INTEGER	Days consumed by approved requests

leave_requests

Individual leave applications.

Column	Type	Description
user_id / leave_type_id	INTEGER FK	Employee and leave type
start_date / end_date	TEXT	YYYY-MM-DD range
days	INTEGER	Actual working days selected
dates_json	TEXT	JSON array of exact selected dates
status	TEXT	pending approved rejected
approved_by / approved_at	INTEGER / TEXT	Who approved and when

app_settings

Key-value store for system configuration.

Column	Type	Description
key	TEXT PK	Setting name
value	TEXT	smtp_host, smtp_port, smtp_user, smtp_pass, smtp_from, email_enabled

5. User Roles & Permissions

Feature	Employee	Manager	HR Admin
View own dashboard	✓	✓	✓
Punch in / Punch out	✓	✓	✓
View own attendance	✓	✓	✓
Apply for leave	✓	✓	✓
Download personal PDF report	✓	✓	✓
View team attendance	—	✓	✓
Approve / reject leave	—	✓	✓
Manual clock-out for team	—	✓	✓
Download team Excel/PDF report	—	✓	✓
Manage employees	—	—	✓
Manage branches & geofence	—	—	✓
View all company attendance	—	—	✓
Download company-wide report	—	—	✓
Configure email settings	—	—	✓
Toggle geofencing on/off	—	—	✓

6. Feature Reference

6.1 Authentication & Password Reset

Session-based authentication using Flask server-side sessions. Passwords stored as SHA-256 hashes. Password reset uses a secure URL-safe token valid for 1 hour.

Method	Endpoint	Description
POST	/api/login	Validate email + SHA-256 password, create session
POST	/api/logout	Clear server session
POST	/api/forgot-password	Generate reset token (1hr expiry), return demo token
POST	/api/reset-password	Validate token, update hashed password

- In production, the reset token should be emailed via SMTP rather than returned in the API response.
- Sessions are signed with SECRET_KEY environment variable.

6.2 Attendance — Punch In/Out

Employees clock in and out from the dashboard. Status is determined by comparing punch time to shift_start + 15 minute grace period. GPS coordinates are captured and stored.

Method	Endpoint	Description
POST	/api/punch-in	Record clock-in with GPS, determine on-time vs late
POST	/api/punch-out	Record clock-out with GPS coordinates
GET	/api/attendance/today	Get today attendance record for logged-in user
GET	/api/attendance/me?month=	Monthly attendance history for logged-in user
GET	/api/attendance/summary	Monthly counts: on-time, late, absent, leave
GET	/api/attendance/team?date=	Team attendance for a specific date (Manager/HR)

- On-time: punched in $\leq$ shift_start + 15 minutes.
- Late: punched in $>$ shift_start + 15 minutes.
- Location label (e.g. "47m from Head Office") stored in geo_in / geo_out fields.

6.3 Geofencing

Each punch in/out validates the employee GPS position against their assigned branch coordinates. Distance calculated using the Haversine formula. Default radius is 200 metres.

Method	Endpoint	Description
POST	/api/branches/save	Create or update a branch with GPS coords and radius
GET	/api/branches	List all branches
POST	/api/branches/delete	Delete a branch

- Geofencing can be disabled globally via Settings → Geofence toggle.

- If an employee has no branch assigned, the first branch in the database is used as fallback.
- HR Admin can use "Use my current location" in the branch form to auto-fill coordinates.

6.4 Leave Management

Employees select individual dates on an interactive calendar. The system counts only selected working days and stores the exact date list (dates_json) to prevent range miscounting.

Method	Endpoint	Description
GET	/api/leave/types	List available leave types
GET	/api/leave/balance	Employee leave balances for current year
POST	/api/leave/apply	Submit leave request with selected dates array
GET	/api/leave/my-requests	Logged-in user leave request history
GET	/api/leave/pending	Pending requests for manager/HR approval
POST	/api/leave/action	Approve or reject a leave request

- Leave types: Annual (14d), Sick (14d), Emergency (3d), Maternity (90d), Paternity (7d), Unpaid (30d).
- Calendar blocks weekends, past dates, Indonesian national holidays (2025–2026), and already-applied dates.
- Hovering over a holiday shows its name as a tooltip.
- When submitted, supervisor receives an approval email (if SMTP configured).
- Approved leave auto-marks attendance records as "On Leave" for the exact selected dates.

6.5 Overtime & Auto Clock-Out

A two-phase timer system runs in the browser every second on the dashboard.

Method	Endpoint	Description
POST	/api/overtime/set	Store planned overtime checkout time
POST	/api/overtime/auto-checkout	System auto clock-out at planned time
POST	/api/overtime/missing-report	Email supervisor about missing OT clock-out
POST	/api/overtime/manual-checkout	Manager manually enters clock-out time
GET	/api/attendance/missing-checkouts?date=	List employees missing clock-out

- Regular phase: Warning popup at shift_end – 15 min asking if doing overtime. If no response, silent auto clock-out at shift_end. No email.
- Overtime phase: Employee enters extra hours (e.g. 2h → new checkout 19:00). Warning popup at new checkout – 15 min.
- If employee ignores overtime warning → supervisor notified by email (money involved).
- Manager team page shows red "Missing Clock-Out" panel with manual time entry.

6.6 Reporting

All report generation is client-side. PDF uses browser window.print(). Excel uses SheetJS library.

Method	Endpoint	Description
GET	/api/reports/my-attendance?month=	Personal attendance + leave data for PDF

Method	Endpoint	Description
GET	/api/reports/team-attendance?month=&dept=Team/company	Team/company data for Excel/PDF

- Employee: personal PDF with summary stats, attendance table, leave requests.
- Manager: team attendance + leave Excel (3 sheets: Attendance, Leave, Summary) and PDF.
- HR Admin: company-wide or per-department Excel/PDF with department filter dropdown.
- Excel file contains 3 sheets: Attendance Detail, Leave Requests, Summary by Employee.

6.7 Email Notifications

Gmail SMTP integration for automated notifications. Uses Python smtplib with STARTTLS.

Method	Endpoint	Description
POST	/api/settings/save	Save SMTP credentials and settings
POST	/api/settings/test-email	Send a test email to verify configuration

- Leave submitted → supervisor receives formatted HTML email with dates, type, days, reason.
- Missing overtime clock-out → supervisor notified with clock-in time.
- Password reset → token returned in API (in production: email the token).
- Configure via: HR Admin → Settings → Gmail Address + App Password.
- Get Gmail App Password at: myaccount.google.com/apppasswords (requires 2FA enabled).

6.8 Branch & Geofence Management

HR Admin can create multiple office branches, each with a name, address, GPS coordinates, and allowed radius. Employees are assigned to a branch and validated against it on punch.

- Supports multiple branches for companies with offices in different locations.
- "Use my current location" button auto-fills GPS from browser.
- Radius range: 50m – 5000m (default 200m).
- GPS coordinates obtainable from Google Maps: right-click any location → "What's here?"

6.9 Employee Management

HR Admin manages the full employee roster. Each employee stores first/last name separately, department, assigned branch, supervisor, and shift hours.

Method	Endpoint	Description
GET	/api/users	List all employees with manager and branch names
POST	/api/users/add	Create new employee with leave balances
POST	/api/users/update	Update employee details

- New employees automatically get leave balances for the current year.
- Supervisor assignment determines who receives leave approval emails.
- Edit button on each employee row opens a pre-filled edit modal.

7. Version History

v1.0	Initial Release	April 2026
• Login page with forgot/reset password • Employee dashboard with punch in/out • Attendance history with monthly calendar view • Leave application with date range picker • Leave approval workflow (Manager/HR) • Leave balance tracking per leave type • Team attendance view for managers • Employee management (HR Admin) • 3 user roles: Employee, Manager, HR Admin • SQLite database (initial)		
v1.1	Calendar Leave Picker	April 2026
• Replaced date range inputs with interactive click-to-select calendar • Multi-date individual day selection (non-consecutive days supported) • dates_json column added to store exact selected dates • Fixed day count calculation (range vs actual selected days bug) • Calendar shows already-applied dates as blocked		
v1.2	Layout & UX Fixes	April 2026
• Fixed modal height overflow — header/footer always visible • Calendar grid uses flex layout to fill available space • Compact modal spacing for leave form • Fixed sidebar grid layout and content overflow		
v1.3	Geofencing & Multi-Branch	April 2026
• GPS geolocation on punch in/out (Haversine formula) • Branches table: name, address, GPS coordinates, radius • Each employee assigned to a branch • Geofence validation on every punch (403 if outside radius) • Location label stored in attendance (geo_in / geo_out) • Location column added to Team Attendance table • "Use my current location" button in branch form		
v1.4	Employee Form & Email	April 2026
• First name / last name fields (stored separately) • Supervisor dropdown (links employee to correct approver) • Branch dropdown in employee form • Edit employee modal with pre-filled fields • Gmail SMTP integration (App Password) • Leave approval email to supervisor with HTML template • Missing clock-out email notification • Test email button in Settings		

- Branches & Geofence page (HR Admin)
- Settings page (HR Admin)
- Toast notification system

v1.5 PostgreSQL & Railway Deploy

April 2026

- Migrated from SQLite to PostgreSQL (psycopg2-binary)
- All queries updated from ? to %s placeholders
- Auto DB initialization on module load (gunicorn compatible)
- Procfile + railway.json for Railway deployment
- .python-version file (Python 3.11.9 for psycopg2 compatibility)
- /setup-db-workpulse-2026 browser-accessible init endpoint
- /clear-today-workpulse-2026 endpoint for testing
- SECRET_KEY and DATABASE_URL environment variables

v1.6 Jakarta Timezone & Holidays

April 2026

- TZ environment variable support (default: Asia/Jakarta)
- All datetime.now() and date.today() use zoneinfo timezone
- Indonesia national holidays 2025–2026 built into leave calendar
- Holiday dates shown in orange on calendar (non-clickable)
- Flag emoji (🇮🇩) on holiday dates with tooltip name
- Holiday legend item added to calendar

v1.7 Overtime & Auto Clock-Out

April 2026

- Two-phase browser timer system (checks every second on dashboard)
- Regular phase: silent auto clock-out at shift end (no email)
- Shift-end popup at shift_end – 15min: "Are you doing overtime?"
- Overtime phase: employee enters extra hours, new checkout set
- Overtime popup at OT_end – 15min with supervisor notification if ignored
- performAutoCheckout() API call with optional supervisor notification
- Missing clock-out panel on Team Attendance page
- Manual clock-out entry for managers
- /api/overtime/* backend routes
- /api/attendance/missing-checkouts endpoint

v1.8 Reporting

April 2026

- Reports page accessible to all roles (■ in sidebar)
- Employee: personal attendance + leave PDF (browser print)
- Manager: team attendance + leave Excel (.xlsx, 3 sheets) and PDF
- HR Admin: company-wide or per-department Excel/PDF
- Department filter for HR company reports
- Report type selector: Attendance / Leave / Both
- SheetJS (xlsx 0.18.5) loaded from CDN for Excel generation
- /api/reports/my-attendance and /api/reports/team-attendance endpoints
- Summary by employee section in all team reports

8. Deployment Guide (Railway)

Step 1	Create GitHub Repository Create a private GitHub repository named workpulse. Upload all project files (app.py, static/, templates/, requirements.txt, Procfile, railway.json, .python-version, README.md).
Step 2	Create Railway Account Sign up at railway.app using GitHub login. No credit card required. Free tier includes \$5 credit/month.
Step 3	Create PostgreSQL Database In Railway dashboard: New → Database → PostgreSQL. Railway provisions a managed database instantly.
Step 4	Deploy Web Service New → Web Service → Connect GitHub repo (workpulse). Railway auto-detects Python and uses Procfile.
Step 5	Configure Environment Variables In web service Variables tab: add DATABASE_URL (reference from Postgres service) and SECRET_KEY (any random string).
Step 6	Set Timezone Add TZ = Asia/Jakarta (or Asia/Makassar / Asia/Jayapura) as environment variable.
Step 7	Initialize Database After first deploy, visit: https://your-app.railway.app/setup-db-workpulse-2026 to create tables and seed demo data.
Step 8	Generate Public URL Settings → Networking → the URL is already generated. Share with employees.
Step 9	Configure Email (optional) Login as HR Admin → Settings → enter Gmail address and App Password. Click Test Email to verify.
Step 10	Set Branch Coordinates HR Admin → Branches & Geofence → edit Head Office with real GPS coordinates from Google Maps.

9. Environment Variables

Variable	Required	Default	Description
DATABASE_URL	Yes	—	PostgreSQL connection string (postgres://... or postgresql://...)
SECRET_KEY	Yes	random	Flask session signing key — use a long random string
TZ	Recommended	UTC	Server timezone (e.g. Asia/Jakarta, Asia/Makassar, Asia/Jayapura)
PORT	Auto	5000	Set automatically by Railway; Gunicorn binds to this port

■■ Never commit SECRET_KEY or smtp_pass to GitHub. Store all sensitive values as Railway environment variables only.

10. Demo Accounts

The following accounts are seeded automatically when /setup-db-workpulse-2026 is called. Change all passwords immediately in production.

Role	Name	Email	Password	Department
HR Admin	Sarah Johnson	hr@company.com	hr123	HR
Manager	David Chen	manager@company.com	mgr123	Engineering
Employee	Alice Wong	alice@company.com	emp123	Engineering
Employee	Bob Martinez	bob@company.com	bob123	Engineering
Employee	Carol Kim	carol@company.com	carol123	Design

Leave Types (seeded)

Leave Type	Max Days/Year	Notes
Annual Leave	14 days	Standard annual paid leave
Sick Leave	14 days	Medical leave with documentation
Emergency Leave	3 days	Family emergency or bereavement
Maternity Leave	90 days	Birth mother maternity leave
Paternity Leave	7 days	Father paternity leave
Unpaid Leave	30 days	Extended leave without pay

API Endpoint Summary

Method	Endpoint	Access	Description
POST	/api/login	All	Authenticate user
POST	/api/logout	All	End session
POST	/api/forgot-password	Public	Request password reset token
POST	/api/reset-password	Public	Reset password with token
POST	/api/punch-in	All	Clock in with GPS
POST	/api/punch-out	All	Clock out with GPS
GET	/api/attendance/today	All	Today attendance record
GET	/api/attendance/me	All	Monthly attendance history
GET	/api/attendance/summary	All	Monthly attendance counts
GET	/api/attendance/team	Mgr/HR	Team attendance by date
GET	/api/attendance/missing-checkouts	Mgr/HR	Missing clock-outs
GET	/api/branches	All	List all branches
POST	/api/branches/save	HR	Create/update branch
POST	/api/branches/delete	HR	Delete branch

Method	Endpoint	Access	Description
GET	/api/settings	HR	Get system settings
POST	/api/settings/save	HR	Save system settings
POST	/api/settings/test-email	HR	Send test email
GET	/api/leave/types	All	List leave types
GET	/api/leave/balance	All	User leave balances
POST	/api/leave/apply	All	Submit leave request
GET	/api/leave/my-requests	All	User leave history
GET	/api/leave/pending	Mgr/HR	Pending leave approvals
POST	/api/leave/action	Mgr/HR	Approve or reject leave
GET	/api/users	HR	List all employees
POST	/api/users/add	HR	Create new employee
POST	/api/users/update	HR	Update employee details
POST	/api/overtime/set	All	Store OT planned checkout
POST	/api/overtime/auto-checkout	All	System auto clock-out
POST	/api/overtime/missing-report	All	Email supervisor re: missing OT
POST	/api/overtime/manual-checkout	Mgr/HR	Manual clock-out entry
GET	/api/reports/my-attendance	All	Personal report data
GET	/api/reports/team-attendance	Mgr/HR	Team/company report data